Hyak Klone Onboarding
CHEME 498C/599A

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INFORMATION TECHNOLOGY
UNIVERSITY of WASHINGTON

Hyak Klone

UW's Supercomputer

- HPC Cluster
- Condo Model
- Storage
- Community Idle Resources

- 44,184 compute cores
- 954 GPUs

USER SUPPORT

Helpdesk

help@uw.edu

- Hyak, Tillicum, Kopah, Lolo, Cloud
- Research Computing Consulting
- Support Request form

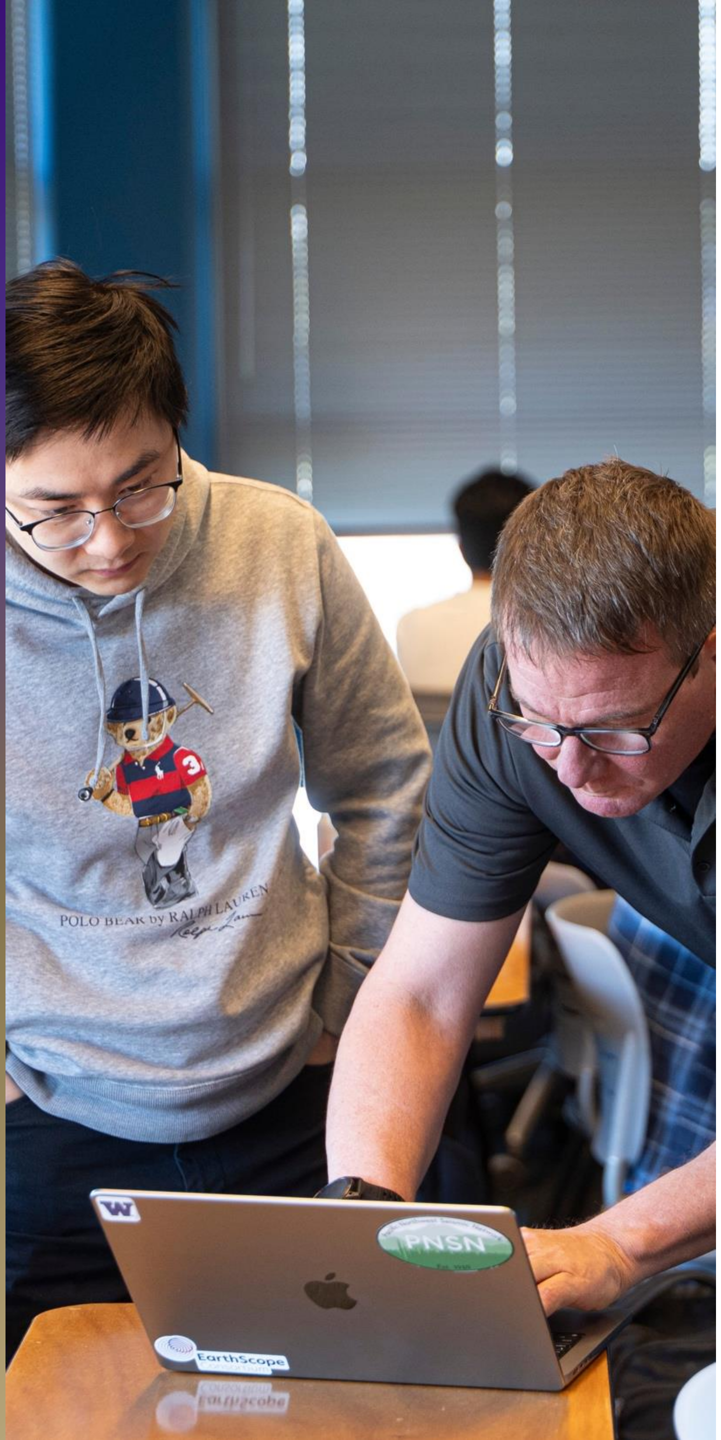
Office Hours

- Drop-in or by Appointment
  - Calendar
- In-person and remote

Regular Training Opportunities

- Recorded
- Events Calendar
- We have a mailing list

Everything underlined is a link



Documentation

www.hyak.uw.edu/

- Step by step guides to tools and software
- Monthly blog posts with updates
- Links to support and services

Learning Resources

www.hyak.uw.edu/learn

- Short How-to videos
- Full length tutorials

Everything underlined is a link

RESEARCH COMPUTING CLUB

STF | Student Technology Fee Committee

Computing Resources

Hyak Klone:

- 26 nodes
- 24 GPUs
- Priority job scheduling
- Computing storage

Tillicum:

- Apply for $250 in Tillicum computing credits

Cloud Credits Program

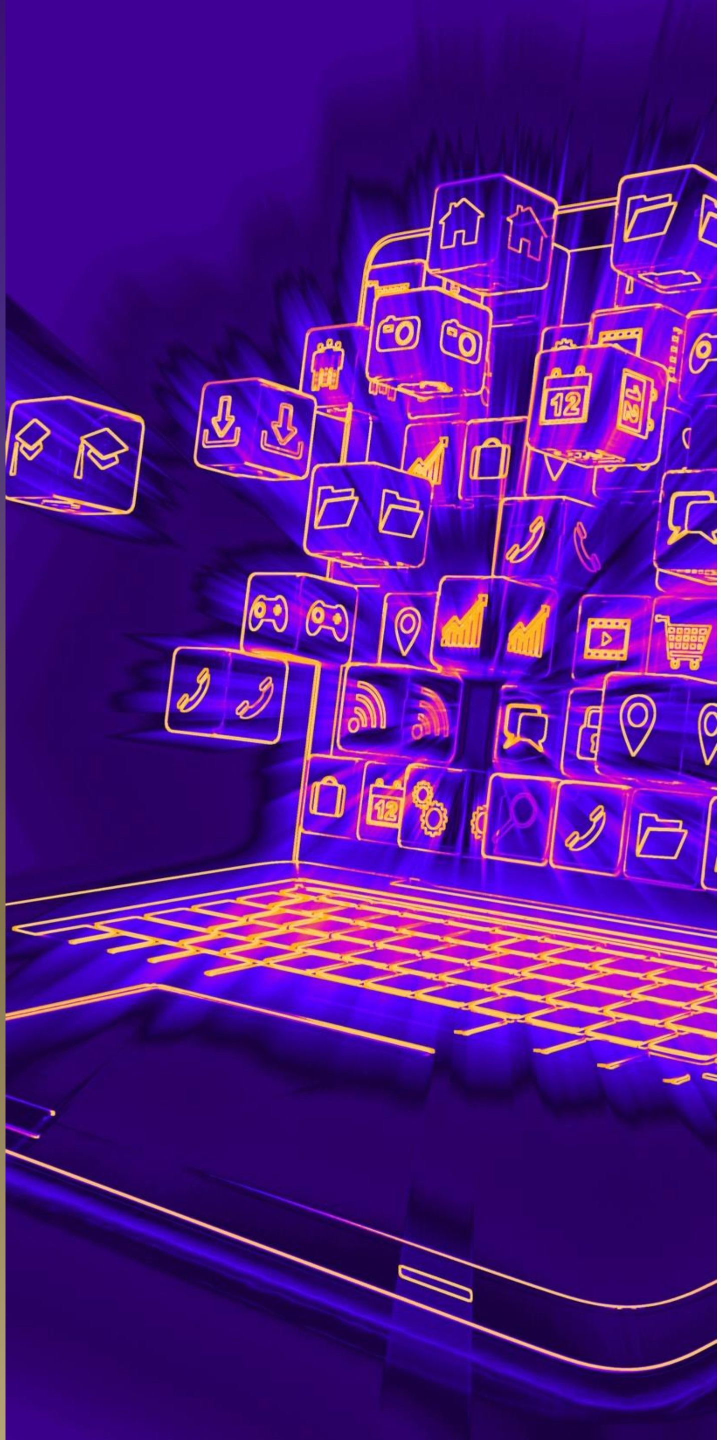
- Apply for $500 in AWS credits
- Guidance and troubleshooting

Slack Community

- Get help from your peers
- Get notified about events, jobs, and fellowships

Special Events

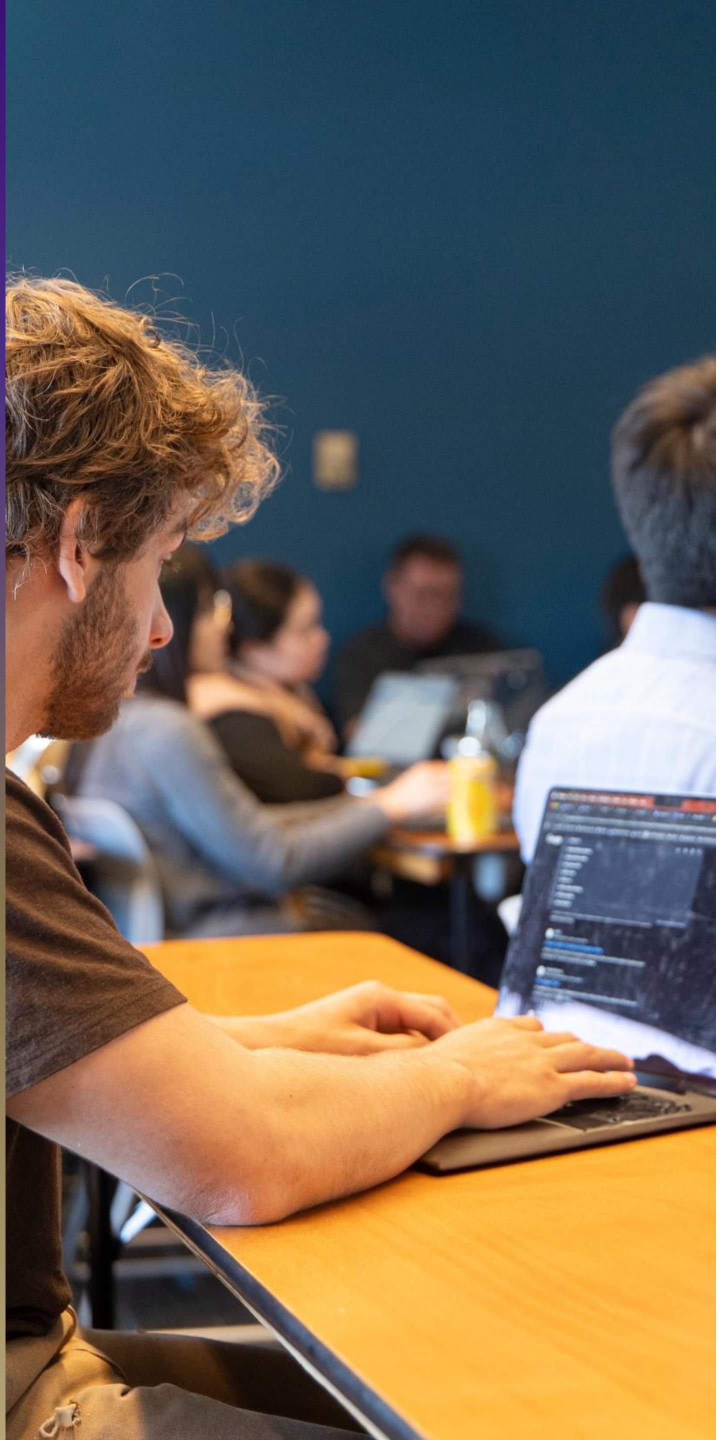
- Hackathon
- AWS GameDay



Need To Know

Software Policies

- Shared Research Environment
- Baseline Computing Environment
- No Root/Sudo Access
- Researcher Responsibility
- Hyak Team - Support, Training, and Documentation



Need To Know

Environment – Rocky Linux + Slurm

- Open OnDemand (GUI) - Jupyter Notebooks
- Containerized environments - Apptainer runtime
- LMOD modules - maintained and contributed modules
- Python environments - Conda module

Logging in to Klone

Log in to Klone using SSH:

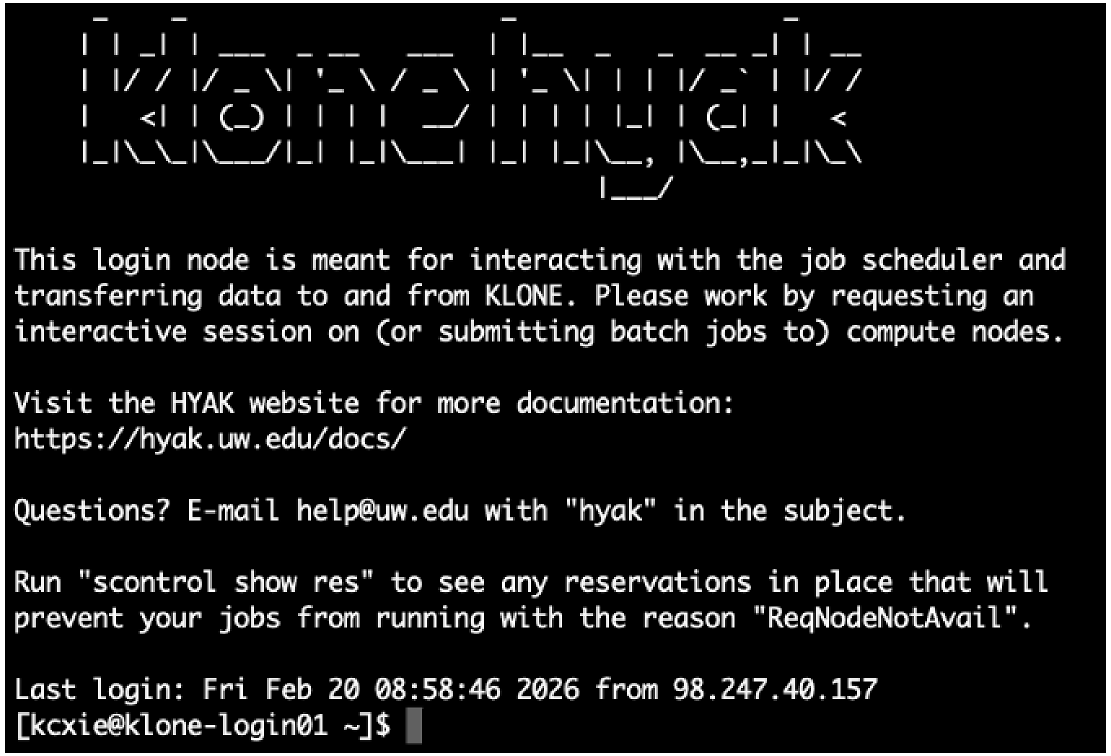
- Set up two-factor authentication
- Use SSH to log into Klone

- Mac and Linux Users: Open the Terminal

- Windows Users: PowerShell / WSL / Git Bash

- Command:

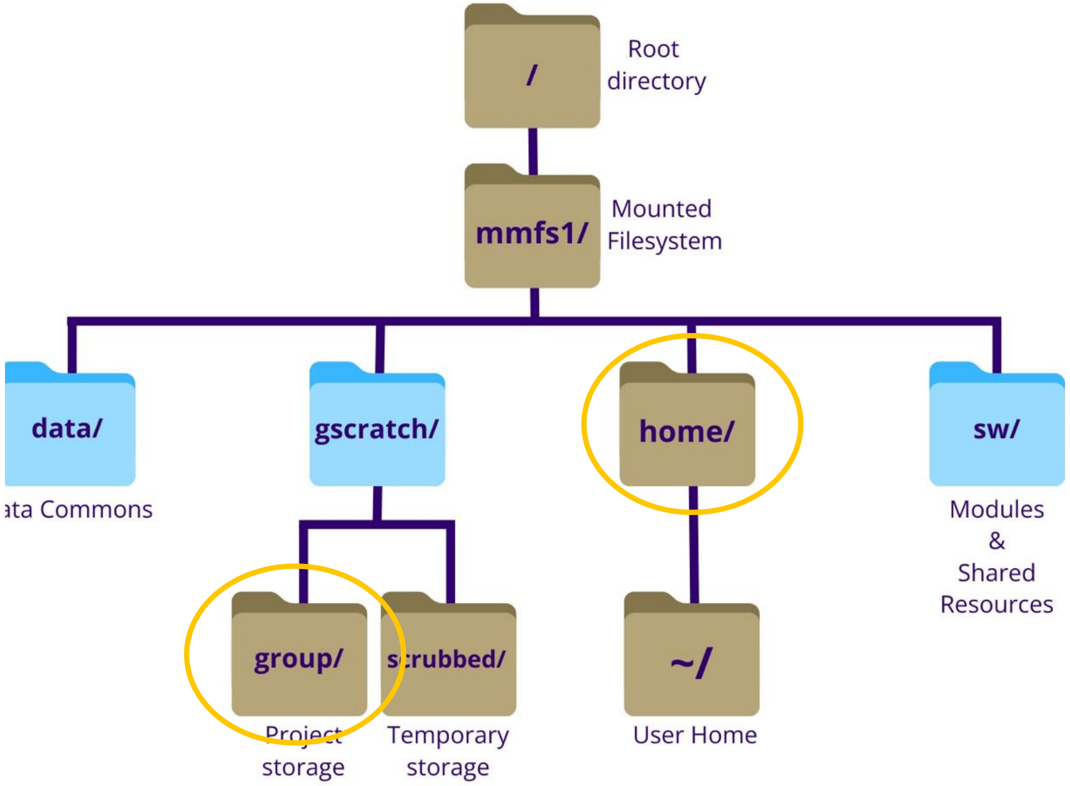
ssh UWNetID@klone.hyak.uw.edu



Filesystem and Storage on Klone

Storage spaces for Klone users:

- Home directory (/mmfs1/home/UWNetID): 10GB, personal
- Lab/project dedicated storage (/mmfs1/gscratch/<lab>): shared storage for labs/projects



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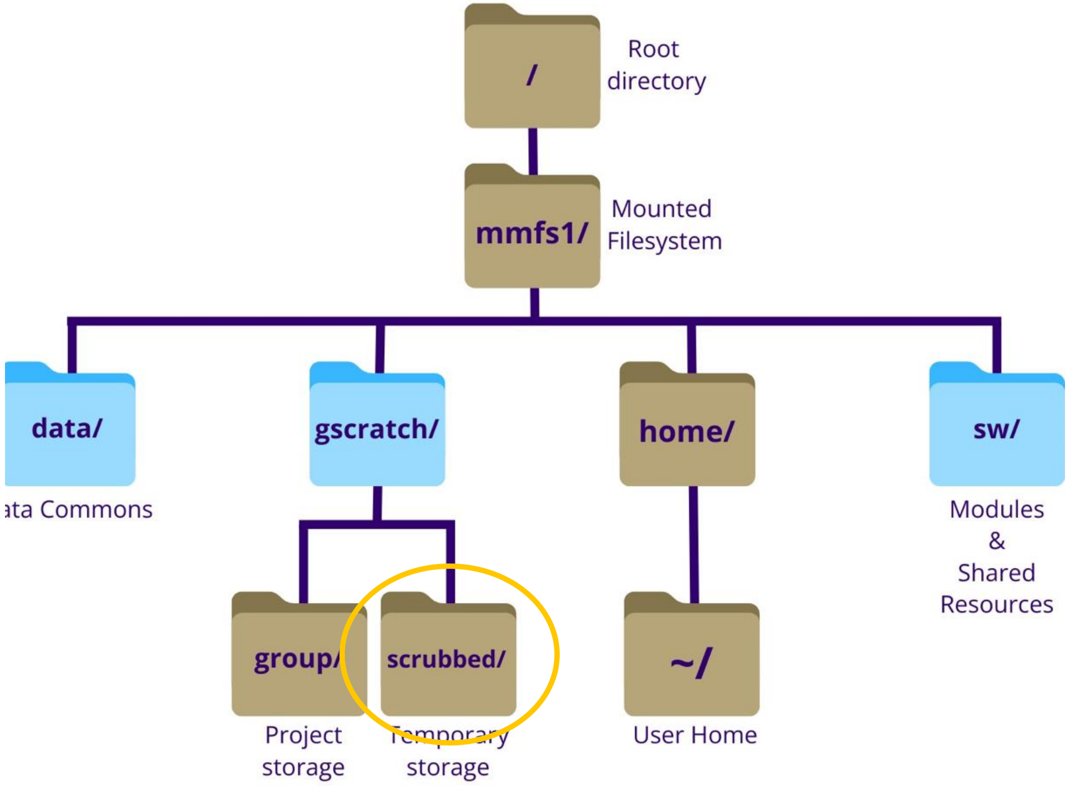
- Scrubbed storage (/mmfs1/gscratch/scrubbed): large, temporary scratch space for active computation



Learning Objectives

- Log in and access Hyak Klone resources
- Navigate the filesystem and manage data
- Transfer data with scp and Globus
- Use essential Linux commands for inspecting and manipulating files
- Submit and monitor jobs with Slurm
- Load and manage software modules and computing environments

Training Materials Repo

https://github.com/UWrc/klone-onboarding-cheme599a.git

Everything underlined is a link

Learn more – everything underlined is a link

UWIT - https://it.uw.edu/

Research Computing Services

- Tillicum: GPU-Accelerated Research Computing Platform
  - Intake Form
- Hyak: High-Performance Supercomputing Research Cluster
  - Pricing and Eligibility
  - Documentation
- Data Storage Services
  - Kopah S3 Object Storage
    - Pricing and Eligibility (cost calculator)
    - Documentation
  - Lolo Data Archive
    - Pricing and Eligibility
    - Documentation
- Cloud Computing
- Computing for Restricted Access Data
- Research Computing Consulting

Training and Events

- Hyak mailing list
- Hyak Blog
- UWIT Research Computing Calendar
- Office Hours
- eScience Newsletter (scroll for sign up)
- eScience Data Science and AI Accelerator
- Office of Research Calendar

Past Trainings

- Research Computing Tutorials & Video Library
- eScience YouTube Channel

New Hyak Users

- Free Demonstration Account
- Hyak Basics Tutorial
- Limitations of demonstration accounts
- Using free resources (Checkpoint)

Students

- Research Computing Club
- Student Hyak account
- Cloud Credits
- Student Technology Fee

Other Resources

- **Join the UW AI Community of Practice** on MS Teams to get updates from UW-IT's AI team about events and join the discussion around AI in the news, society, and culture.
- UW Seattle Events Calendar

Special thanks to

- UW Office of Research
- eScience Institute